

# On common $\mathcal{M}$ -subspaces of quadratic (vectorial) Boolean bent functions

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An  $(n, k)$ -*Boolean function* is a function from  $\mathbb{F}_2^n$  to  $\mathbb{F}_2^k$   
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Any vectorial  $(n, k)$ -function  $F$  (i.e.,  $k \geq 2$ ) can be written as  $F(x) = (f_1(x), \dots, f_k(x))$ , where the Boolean functions  $f_i$  on  $\mathbb{F}_2^n$  are called *coordinate functions* of  $F$ , and their linear combinations are called its *component functions*.

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The *derivative* of an  $(n, k)$ -Boolean function  $f$  at  $a \in \mathbb{F}_2^n$  is the function defined by

$$D_a F(x) = F(x + a) + F(x),$$

and the  $k$ -th order derivative of  $F$  at  $v_1, v_2, \dots, v_k \in \mathbb{F}_2^n$  is defined recursively by

$$D_{v_1} D_{v_2} \dots D_{v_k} F = D_{v_1} (D_{v_2} \dots D_{v_k} F).$$

An  $(n, k)$ -Boolean function  $F$  is  
affine if all its first order derivatives  $D_a F$  are constant,  
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Or, via ANF:

Any Boolean function  $f$  can be uniquely represented by its algebraic normal form as a polynomial in  $\mathbb{F}_2[x_1, \dots, x_n]/(x_1 + x_1^2, \dots, x_n + x_n^2)$

$$f(x_1, \dots, x_n) = \sum_{a=(a_1, \dots, a_n) \in \mathbb{F}_2^n} \mu_a x_1^{a_1} \cdots x_n^{a_n},$$

where  $\mu_a \in \mathbb{F}_2$ , for all  $a = (a_1, \dots, a_n) \in \mathbb{F}_2^n$ .

For a vector  $x \in \mathbb{F}_2^n$ , the *Hamming weight* of  $x$ , denoted by  $\text{wt}(x)$ , is the number of its nonzero entries. The Hamming weight of a Boolean function  $f$  is  $\text{wt}(f) = |\text{supp}(f)| = |\{x \in \mathbb{F}_2^n : f(x) \neq 0\}|$ , and the Hamming distance between two Boolean functions  $f$  and  $g$  is defined as  $d(f, g) = \text{wt}(f + g)$ .

An  $(n, k)$ -Boolean function is balanced if all its component functions have weight  $2^{n-1}$  (*ignoring the trivial component function*).

$F$  is **bent** if  $D_a F$  is balanced for all  $a \in \mathbb{F}_2^n \setminus \{0_n\}$ .

*The Walsh-Hadamard transform* of an  $n$ -variable Boolean function  $f$  is the mapping  $W_f : \mathbb{F}_2^n \rightarrow \mathbb{Z}$ , defined by:

$$W_f(w) = \sum_{x \in \mathbb{F}_2^n} (-1)^{f(x) + x \cdot w}, \text{ for every } w \in \mathbb{F}_2^n.$$

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The *nonlinearity*  $nl(f)$ , is the minimum Hamming distance between  $f$  and the set of affine functions (in the same number of variables).

$$\begin{aligned} W_f(a) &= |\{x \in \mathbb{F}_2^n : f(x) = l_a(x)\}| - |\{x \in \mathbb{F}_2^n : f(x) \neq l_a(x)\}| \\ &= 2^n - 2|\{x \in \mathbb{F}_2^n : f(x) \neq l_a(x)\}| = 2^n - 2d(f, l_a), \end{aligned}$$

hence  $d(f, l_a) = 2^{n-1} - \frac{1}{2} W_f(a)$  and

$$nl(f) = 2^{n-1} - \frac{1}{2} \max_{a \in \mathbb{F}_2^n} |W_f(a)|.$$

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$$nl(f) \leq 2^{n-1} - \frac{1}{2} 2^{n/2}$$

and

$$nl(f) = 2^{n-1} - \frac{1}{2} 2^{n/2} \text{ iff } |W_f(a)| = 2^{n/2} \text{ for all } a \in \mathbb{F}_2^n \text{ iff } f \text{ is bent.}$$

$f$  is bent if and only if  $|W_f(a)| = 2^{n/2}$  for all  $a \in \mathbb{F}_2^n$ .

Restrictions on affine subspaces:

Let  $M = \max_{v \in \mathbb{F}_2^n} |W_f(v)|$ , and let  $V$  be a subspace of  $\mathbb{F}_2^n$ .

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If  $|V| = M$ , then

$$f(v) = a \cdot v + b, \text{ for } v \in c + V,$$

if and only if

$$W_f(w) = (-1)^{c \cdot (a+w)+b} M, \text{ for } w \in a + V^\perp.$$

The *Maiorana-McFarland class*  $\mathcal{M}$  is the set of  $n$ -variable Boolean functions of the form

$$f(x, y) = x \cdot \pi(y) + g(y), \text{ for all } x, y \in \mathbb{F}_2^{n/2},$$

where  $\pi$  is a permutation of  $\mathbb{F}_2^{n/2}$ , and  $g$  is an arbitrary Boolean function on  $\mathbb{F}_2^{n/2}$ .

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A vector subspace  $U$  of  $\mathbb{F}_2^n$  is an  $\mathcal{M}$ -*subspace* of an  $(n, m)$ -function  $F$ , if for all  $a, b \in U$  the second-order derivatives satisfy

$$D_a D_b F(x) = F(x) + F(x + a) + F(x + b) + F(x + a + b) = 0_m,$$

for all  $x \in \mathbb{F}_2^n$ .

An  $n$ -variable Boolean bent function  $f$  belongs to  $\mathcal{M}^\#$  if and only if it has an  $\mathcal{M}$ -subspace of dimension  $\frac{n}{2}$ .

For a vectorial Boolean function  $F = (f_1, \dots, f_m)$  a subspace  $V$  of  $\mathbb{F}_2^n$  is an  $\mathcal{M}$ -subspace of  $F$  if and only if  $V$  is an  $\mathcal{M}$ -subspace of each coordinate (component) function  $f_i$ ,  $i = 1, \dots, m$ .

Multiple approaches to extending the Boolean  $\mathcal{M}$ -construction to the vectorial case.

The *Strict Maiorana-McFarland class*  $\mathcal{SM}$ :

$$F(x, y) = L(x\pi(y)) + G(y),$$

where  $\pi$  is a permutation of  $\mathbb{F}_{2^{n/2}}$ ,  $L$  is affine from  $\mathbb{F}_{2^{n/2}}$  onto  $\mathbb{F}_2^m$ , and  $G$  is an arbitrary  $(n/2, m)$ -function.

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The *Extended Maiorana-McFarland class*  $\mathcal{EM}$ :

$$F(x, y) = \psi(x, y) + G(y),$$

where  $G: \mathbb{F}_2^{n/2} \rightarrow \mathbb{F}_2^m$  is arbitrary and  $\psi: \mathbb{F}_2^{n/2} \times \mathbb{F}_2^{n/2} \rightarrow \mathbb{F}_2^m$  satisfies that, for every  $y$ ,  $x \mapsto \psi(x, y)$  is linear and, for every  $x \neq 0$ ,  $y \mapsto \psi(x, y)$  is balanced.

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The *General Maiorana-McFarland class*  $\mathcal{GM}$  is the set of  $(n, m)$ -functions  $F$  such that every (nontrivial) component function belongs to  $\mathcal{M}^\#$ .

## Theorem

Let  $n = 2k$  and  $m \leq k$ . The extended Maiorana-McFarland class  $\mathcal{EM}$  coincides with the class of vectorial  $(n, m)$ -bent functions  $F$  of the form  $F = (f_1, \dots, f_m)$  where

$$f_i(x, y) = x \cdot \pi_i(y) + h_i(y),$$

for some  $h_i \in \mathcal{B}_k$  and  $\pi_i: \mathbb{F}_2^k \rightarrow \mathbb{F}_2^k$  such that  $\langle \pi_1, \dots, \pi_m \rangle$  is an  $m$ -dimensional  $\mathbb{F}_2$ -vector space of permutations.

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## Corollary

$F$  belongs to  $\mathcal{EM}^\#$  if and only if  $F$  has at least one maximal  $\mathcal{M}$ -subspace (shared by all coordinate functions).

Using the known classifications for  $n = 4, 6, 8$ , we verified that every quadratic vectorial  $(n, 2)$ -bent function admits a maximal  $\mathcal{M}$ -subspace:

$n = 4$ : there is a unique EA-class admitting 5 maximal  $\mathcal{M}$ -subspaces,

$n = 6$ : there is a unique EA-class admitting 9 maximal  $\mathcal{M}$ -subspaces,

$n = 8$  there are three EA-classes with 17, 21, and 85 maximal  $\mathcal{M}$ -subspaces.

## Lemma

*Let  $f$  and  $g$  be two quadratic Boolean functions in  $n$  variables. Let  $U$  be a common  $\mathcal{M}$ -subspace of  $f$  and  $g$  with  $k = \dim(U) < \frac{n}{3}$ . Then, there exist at least  $2^{n-3k} - 1$  different subspaces  $V \subset \mathbb{F}_2^n$  such that  $U \subset V$ ,  $\dim(V) = k + 1$  and  $V$  is a common  $\mathcal{M}$ -subspace of  $f$  and  $g$ .*

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Let  $\{u_1, \dots, u_k\}$  be a basis for  $U$  and extend it to a basis

$$\{u_1, \dots, u_k, b_1, \dots, b_{n-k}\} \text{ for } \mathbb{F}_2^n.$$

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**Step 1.** Remove at most one direction  $b_i$  such that  $D_{b_i} D_{u_1}(f + g) = 1$  to obtain  $\{u_1, \dots, u_k, b'_1, \dots, b'_{n-k-1}\}$  with  $D_{b'_j} D_{u_1} f = D_{b'_j} D_{u_1} g$ , by setting  $b'_j = b_i + b_j$  when necessary.

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**Step 2.** Remove at most one direction  $b'_j$  such that  $D_{b'_j} D_{u_1}(f) = 1$  to obtain  $\{u_1, \dots, u_k, b''_1, \dots, b''_{n-k-2}\}$  with  $D_{b''_l} D_{u_1} f = D_{b''_l} D_{u_1} g = 0$ , by setting  $b''_l = b'_j + b'_l$  when necessary.

## Theorem

Let  $f$  and  $g$  be two quadratic Boolean functions in  $n$  variables. Denote by  $\mathcal{M}_t(f, g)$  the set of common  $t$ -dimensional  $\mathcal{M}$ -subspaces of  $f$  and  $g$ .

Then,

$$|\mathcal{M}_t(f, g)| \geq \prod_{k=0}^{t-1} \frac{2^{n-3k} - 1}{2^{k+1} - 1},$$

for all  $t \in \{1, 2, \dots, \lfloor \frac{n-1}{3} \rfloor + 1\}$ .

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Assume the theorem is true for  $t \leq \lfloor \frac{n-1}{3} \rfloor$ .

Since  $t \leq \lfloor \frac{n-1}{3} \rfloor < \frac{n}{3}$ , for every  $U \in \mathcal{M}_t(f, g)$ , there exist at least  $2^{n-3t} - 1$  different subspaces  $V \in \mathcal{M}_{t+1}(f, g)$  such that  $U \subset V$  (Lemma).

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If  $W \in \mathcal{M}_{t+1}(f, g)$ , then every  $t$ -dimensional subspace of  $W$  is in  $\mathcal{M}_t(f, g)$ , and there is  $2^{t+1} - 1$  of them.

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$$|\mathcal{M}_t(f, g)| \geq \prod_{k=0}^{t-1} \frac{2^{n-3k} - 1}{2^{k+1} - 1},$$

for all  $t \in \{1, 2, \dots, \lfloor \frac{n-1}{3} \rfloor + 1\}$ .

$$\begin{aligned} |\mathcal{M}_{t+1}(f, g)| &\geq \frac{2^{n-3t} - 1}{2^{t+1} - 1} \cdot |\mathcal{M}_t(f, g)| \\ &\geq \frac{2^{n-3t} - 1}{2^{t+1} - 1} \cdot \prod_{k=0}^{t-1} \frac{2^{n-3k} - 1}{2^{k+1} - 1} = \prod_{k=0}^t \frac{2^{n-3k} - 1}{2^{k+1} - 1}. \end{aligned}$$

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Adaptation of a result on alternating bilinear forms by J. L. Alperin (1965).

Trivially true for  $n = 2$ . Assume it is true for all  $k$ ,  $2 \leq k < n$ .

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*Case 1:  $n$  is even.* Let  $V$  be any  $(n-1)$ -dimensional subspace of  $\mathbb{F}_2^n$  and set  $f' = f|_V$  and  $g' = g|_V$ . By the inductive hypothesis, there is a common  $\mathcal{M}$ -subspace  $W \subset V$  of  $f'$  and  $g'$  with

$$\dim(W) = \left\lfloor \frac{(n-1) + 1}{2} \right\rfloor = \frac{n}{2} = \left\lfloor \frac{n+1}{2} \right\rfloor.$$

For all  $w_1, w_2 \in W$  and  $x \in \mathbb{F}_2^n$  we have

$$\begin{aligned} D_{w_1} D_{w_2} f(x) &= D_{w_1} D_{w_2} f(0_n) = D_{w_1} D_{w_2} f'(0_n) = 0, \text{ and} \\ D_{w_1} D_{w_2} g(x) &= D_{w_1} D_{w_2} g(0_n) = D_{w_1} D_{w_2} g'(0_n) = 0. \end{aligned}$$

Case 2:  $n$  is odd.  $f$  is EA-equivalent to

$$x_1x_2 + x_3x_4 + \cdots + x_{2t-1}x_{2t},$$

for some  $1 \leq t \leq \frac{n}{2}$ .

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For  $D_v g(x) = a \cdot x + b$  set  $W_1 = \langle a \rangle^\perp$ .

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For any  $w \in W_1$ , we have

$$D_w D_v(g)(0_n) = D_v(g)(w) + D_v(g)(0_n) = a \cdot w = 0.$$

Consequently,  $D_w D_v g = 0$ , for all  $w \in W_1$ .

Let  $W$  be any  $(n-2)$ -dimensional subspace of  $W_1$  not containing  $v$ .

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$$x_1x_2 + x_3x_4 + \cdots + x_{2t-1}x_{2t},$$

for some  $1 \leq t \leq \frac{n}{2}$ . There is  $v \in \mathbb{F}_2^n$  such that  $D_w D_v f = 0$ , for all  $w \in \mathbb{F}_2^n$ .

For  $D_v g(x) = a \cdot x + b$  set  $W_1 = \langle a \rangle^\perp$ .

For any  $w \in W_1$ , we have

$$D_w D_v(g)(0_n) = D_v(g)(w) + D_v(g)(0_n) = a \cdot w = 0.$$

Consequently,  $D_w D_v g = 0$ , for all  $w \in W_1$ .

Let  $W$  be any  $(n-2)$ -dimensional subspace of  $W_1$  not containing  $v$ .

By the inductive hypothesis for  $f' = f|_W$  and  $g' = g|_W$  there is a common  $\mathcal{M}$ -subspace  $Z \subset W$  of  $f'$  and  $g'$  with  $\dim(Z) = \lfloor \frac{(n-2)+1}{2} \rfloor = \frac{n-1}{2}$ .

## Corollary

*Every quadratic vectorial  $(n, 2)$ -bent function  $F$  has the following form, up to EA-equivalence:*

$$F(x, y) = \begin{pmatrix} x \cdot y + g(y) \\ x \cdot L(y) + h(y) \end{pmatrix}^T, \quad \text{where } x, y \in \mathbb{F}_2^{\frac{n}{2}}, \quad g, h \in \mathcal{B}_{\frac{n}{2}},$$

*with  $\deg(g), \deg(h) \leq 2$ , and  $L$  is a linear complete permutation on  $\mathbb{F}_2^{\frac{n}{2}}$  (i.e., both  $L$  and  $L + \text{id}$  are permutations).*

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## Corollary

*For  $(n, 2)$ -bent quadratic functions  $\mathcal{EM}^\#$  and  $\mathcal{GM}^\#$  are the same.*

$(n, 2)$ -bent:

$n = 4$ : there is a unique EA-class admitting 5 maximal  $\mathcal{M}$ -subspaces,

$n = 6$ : there is a unique EA-class admitting 9 maximal  $\mathcal{M}$ -subspaces,

$n = 8$  there are three EA-classes with 17, 21, and 85 maximal  $\mathcal{M}$ -subspaces.

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## Theorem

*Let  $n \geq 2$  be an even number, and let  $f$  and  $g$  be two quadratic Boolean functions in  $n$  variables. Then,  $f$  and  $g$  share at least  $2^{\frac{n}{2}} + 1$  common  $\frac{n}{2}$ -dimensional  $\mathcal{M}$ -subspaces.*

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Assume that for some  $1 \leq t \leq 2^{\frac{n}{2}}$  we have already constructed

$$\mathcal{A}_t = \{M_1, M_2, \dots, M_t\},$$

where all  $M_i$  are distinct common  $\frac{n}{2}$ -dimensional  $\mathcal{M}$ -subspaces.

Total number of hyperplanes in  $\mathbb{F}_2^n$  is  $2^n - 1$  and each  $M_i$  is contained in exactly  $2^{\frac{n}{2}} - 1$ .

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The number of hyperplanes containing at least one element of  $\mathcal{A}_t$  is at most

$$t(2^{\frac{n}{2}} - 1) < (2^{\frac{n}{2}} + 1)(2^{\frac{n}{2}} - 1) = 2^n - 1.$$

Not all hyperplanes are covered and there exists a hyperplane  $W$  containing none of  $M_1, \dots, M_t$ . Restrict to  $W$  and continue.

What happens in the quadratic  $(n, 3)$ -functions case?

What happens in the quadratic  $(n, 3)$ -functions case?

Consider the following three vectorial bent functions in  $n = 6, 8, 10$  variables

$$\begin{aligned} F: \mathbb{F}_2^6 &\rightarrow \mathbb{F}_2^3, & F(x) &= (f_1(x), f_2(x), f_3(x)), \\ G: \mathbb{F}_2^8 &\rightarrow \mathbb{F}_2^3, & G(x) &= (g_1(x), g_2(x), g_3(x)), \\ H: \mathbb{F}_2^{10} &\rightarrow \mathbb{F}_2^3, & H(x) &= (h_1(x), h_2(x), h_3(x)), \end{aligned}$$

whose coordinate functions are defined by their ANFs in the following way:

- $f_1(x) = x_1x_4 + x_2x_5 + x_3x_6,$
- $f_2(x) = x_1x_5 + x_1x_6 + x_2x_4 + x_2x_5 + x_3x_4,$
- $f_3(x) = x_1x_3 + x_1x_4 + x_2x_6 + x_4x_5,$

- $g_1(x) = x_1x_5 + x_2x_6 + x_3x_7 + x_4x_8,$
- $g_2(x) =$   
 $x_1x_3 + x_1x_4 + x_3x_4 + x_2x_5 + x_4x_5 + x_3x_6 + x_4x_6 + x_1x_7 + x_3x_7 + x_4x_7 + x_2x_8,$
- $g_3(x) =$   
 $x_1x_2 + x_1x_3 + x_1x_4 + x_2x_4 + x_1x_6 + x_2x_6 + x_5x_6 + x_1x_7 + x_2x_7 + x_3x_7 + x_1x_8,$

- $h_1(x) = x_1x_{10} + x_1x_2 + x_{10}x_2 + x_{10}x_3 + x_1x_4 + x_{10}x_4 + x_2x_4 + x_1x_5 + x_3x_5 + x_4x_5 + x_{10}x_6 + x_3x_6 + x_4x_6 + x_5x_6 + x_1x_7 + x_2x_7 + x_3x_7 + x_5x_7 + x_6x_7 + x_{10}x_8 + x_2x_8 + x_3x_8 + x_4x_8 + x_6x_8 + x_7x_8 + x_1x_9 + x_{10}x_9 + x_2x_9 + x_4x_9 + x_5x_9 + x_6x_9,$
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None of the functions  $F$ ,  $G$  and  $H$  admits an  $\mathcal{M}$ -subspace of dimension  $n/2$ .

Consequently,  $F$ ,  $G$  and  $H$  are in  $\mathcal{GM}^\#$  but they are outside of the  $\mathcal{EM}^\#$  class.

## Lemma

*The direct sum  $F(x) + G(y)$  of vectorial bent functions is outside  $\mathcal{EM}^\#$  if and only if at least one of the vectorial bent functions  $F$  or  $G$  is outside  $\mathcal{EM}^\#$ .*

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## Theorem

*For any  $n \geq 6$ , there exist quadratic vectorial  $(n, 3)$ -bent functions in  $\mathcal{GM}^\# \setminus \mathcal{EM}^\#$ .*

## Future work:

- Characterize (or provide a construction of) the quadratic  $(n, 2)$ -vectorial bent functions with exactly  $2^{n/2} + 1$  common  $n/2$ -dimensional  $\mathcal{M}$ -subspaces.
- Extend the results on  $\mathcal{GM}^\# \setminus \mathcal{EM}^\#$  for quadratic  $(n, k)$ -vectorial bent functions for  $k \geq 4$ .
- Provide bounds for the dimensions of  $\mathcal{M}$ -subspaces of quadratic  $(n, k)$ -vectorial bent functions for  $k \geq 3$ .

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# Thank you for your attention.